

an error response to the source). Fairly effective and transparent, packet filters are nonetheless tough to configure. They're also susceptible to IP spoofing wherein packets with forged IP addresses are created to either impersonate an "accepted" source or conceal the identity of the sender.

2. Application gateway: Also referred to as application proxies, they are located between the end user and network. The end user directly contacts the gateway, after which it performs requested function for the user. Specific programs use specific mechanisms. It is however not transparent to users, who must install custom applications to contact the gateways. This type is simple, since it only functions to proxy requests from end users. It also intercepts IP packets from the net. However, it causes strain on system performance.
3. Stateful firewall: It keeps track of the network connections moving across it (TCP streams, for example). It distinguishes legitimate packets for different connections and only allows packets matching a specific connection state. All others will be rejected. Sessions without traffic for a specific period will eventually time out to prevent the table from being filled. Stateful firewalls are advantageous as opposed to packet filters since it need only check the connection against its table rather than an extensive rule-set.
4. Proxy server: "Proxy" meaning "substitute". One of the more popular types, proxy servers act as go-betweens for outside clients seeking information from servers. The request is filed and checked according to various filtering rules hence filtering traffic from certain IP addresses. Direct access to the server is subsequently handled by the proxy on behalf of the client. It may speed up resource management by caching and delivering responses according to specific requests. The servers are hence kept anonymous and safe from attack.

An easy-to-use and powerful firewall can be found in ZoneAlarm, the newest release being ZoneAlarm Security Suite 2009. Besides an inbound intrusion detection system, it can also control which programs can make outbound